MEDOMYCIN – PACKAGE INSERT

1. NAME OF THE MEDICINAL PRODUCT
Medomycin 100mg hard gelatin capsules.

2. QUALITATIVE AND QUANTITATIVE COMPOSITION
Each Medomycin capsule 100mg contains 100mg of doxycycline as doxycycline hyclate.

Excipient with known effect: lactose. Each capsule contains 145.mg lactose (as monohydrate).
For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM
Medomycin hard gelatin capsules are green-blue, sized 2, printed with "Medochemie".

4. CLINICAL PARTICULARS
4.1. Therapeutic indications
Doxycycline is used for the same purposes as tetracyclines.
Doxycycline like all the tetracycline has a wide spectrum of activity and has been used in the treatment of a large number of infections caused by susceptible organisms.

Specific conditions which have responded successfully to doxycycline therapy include the following:
Respiratory infections
Acute bronchitis
Acute exacerbations of chronic bronchitis bronchiectasis, single and multilobar pneumonia
Lung abscess
Upper respiratory tract infections such as pharyngitis and laryngotracheitis ENT
Otitis media, sinusitis, tonsillitis
Genitourinary tract infections (acute and chronic)
Cystitis prostatitis, pyelonephritis, trigonitis, urethritis (gonococcal) and others.
Skin and soft tissue infections abscess acne vulgaris and conglobata, furunculosis, impetigo, infected burns cellulites infected eczema and other infected rashes or dermatoses.
Infected traumatic and post-operative wounds omphalitis and paronychia.

Digestive system infections:
Cholangitis, cholecystitis, petitonitis salmonellosis, shigellosis, typhoid fever, obstetrical and gynaecological infections, acute puerperal infections and pelvic inflammatory disease.

Other infections:
Lymphogranuloma venereum, osteomyelitis, thrombophlebitis. Since doxycycline is a member of the tetracyclines series of antibiotics, it may be expected to be useful in the treatment of infections such as following which respond to other tetracyclines.

Digestive system infections:
Enterocolitis, intestinal amebisis, ophthalmic infections, other infections anthrax, Boutonneuse fever, Brucellosis, Gas-gangrene, Leptospirosis, Listerellosis, Meningococcal meningitis, plague, psittacosis, Q fever, rickettsial pox, Rocky Mountain spotted fever, Sympttis tularemia, typhus, yaws and many others.

Doxycycline has been shown to have in vivo activity against the following micro-organism:
E.coli, Enterococcus, H.influenza, Klebsiella-aerobacter group. Klebsiella pneumonia, Nisseria gonorrhoea, Diplococcus pneumonia, Pseudomonas salmonella, Shigella, Staphylococcus, Betahemolytic streptococcus and as a member of the tetracycline series of antibiotics may be expected to share the tetracyclines spectrum of activity which includes susceptible strains of the following additional organisms;
Actinomyces species, Bacillus anthracis, Bacteroides species, Bartonella bacillifomis, Bordetella pertussis, Brucella species, Clostridium species, Corynebacteria species, Donovinia granulomatis, Francisella tularensis, Hemophilus ducreyl, H. pertussis, Listeria monocytogenes, Mimeaae species, Nisseria meningitides, Pasturella pestis, Spirillium minus, Spirochetes, Streptobacillus moniliformis, Vibrio cholera, Mycobacterium tuberculosis, Mycoplasma

pneumonia, rickettsiae, certain parasites (entamoeba histolytica, enterobius vermicularis) and certain viruses of the lymphogranuloma psittacosis trachoma group.

4.2. Posology and method of administration

The usual dose is the equivalent of 200mg of doxycycline initially, followed by 100mg daily. In the management of more severe infections (particularly chronic infections of the urinary tract) 200mg daily, should be used throughout the treatment period. The daily dose may be administered once a day ortwo equally divided doses.

Acute gonococcal anterior urethritis in males, a single dose of 300mg or 100mg bid for 2 – 4 days. Acute gonococcal infections in the adult female should be treated with doses of 100mg bid until cure is effected.

In the treatment of primary and secondary syphilis the recommended dosage is 200mg daily for at least 10 days. When used in streptococcal infections, therapy should be at least continued for 10 days to prevent the development of rheumatic fever of glomerulonephritis.

Children: the recommended oral dose of doxycycline is 1mg per kg body weight on day one of treatment followed by 5mg per kg body weight on subsequent days of therapy.
In the management of very severe infections, 1mg per kg body weight daily should be used throughout the treatment period.

For children over 45kg, the recommended adult dosage regime should be used. The daily dose may be administered once a day or in two equally divided doses given every 12 hours.

<u>Method of administration</u>
Medomycin capsules should be taken with adequate amounts of fluid (at least 100ml of water). This should be done in the sitting or standing position and the patient should be advised to remain upright for at least thirty minutes after taking a dose. Medomycin capsules should be taken well before bedtime to reduce the risk of oesophageal irritation and ulceration. If gastric irritation occurs, it is recommended that Medomycin be given with food or milk. Studies indicate that the absorption of Medomycin is not notably influenced by simultaneous ingestion of food or milk.

<u>Route of administration</u>
Oral administration

4.3. Contraindications

Hypersensitivity to the active substance or or to any other tetracyclines, or to any of the excipients listed in section 6.1
Obstructive oesophageal disorders, such as stricture or achalasia.
The use of drugs of the tetracycline class during tooth development (pregnancy, infancy and childhood to the age of 12 years) may cause permanent discolouration of the teeth (yellow-grey-brown). This adverse reaction is more common during long-term use of the drugs but has been observed following repeated short-term courses. Enamel hypoplasia has also been reported. Medomycin is therefore contraindicated in these groups of patients.
Pregnancy: Medomycin is contraindicated in pregnancy. It appears that the risks associated with the use of tetracyclines during pregnancy are predominantly due to effects on teeth and skeletal development. (See above about use during tooth development).
Nursing mothers: Tetracyclines are excreted into milk and are therefore contra-indicated in nursing mothers. (See above about use during tooth development).
Children: Medomycin is contraindicated in children under the age of 12 years. As with other tetracyclines, doxycycline forms a stable calcium complex in any bone-forming tissue. A decrease in the fibula growth rate has been observed in prematures given oral tetracyclines in doses of 25mg/kg every 6 hours. This reaction was shown to be reversible when the drug was discontinued. (See above about use during tooth development).

4.4. Special warnings and precautions for use

Use in patients with impaired hepatic function: Medomycin should be administered with caution to patients with hepatic impairment or those receiving potentially hepatotoxic drugs.

Abnormal hepatic function has been reported rarely and has been caused by both the oral and parenteral administration of tetracyclines, including doxycycline.

Use in patients with renal impairment: Excretion of doxycycline by the kidney is about 40%/72 hours in individuals with normal renal function. This percentage excretion may fall to a range as low as 1-5%/72 hours in individuals with severe renal insufficiency (creatinine clearance below 10ml/min). Studies have shown no significant difference in the serum half-life of doxycycline in individuals with normal and severely impaired renal function. Haemodialysis does not alter the serum half-life of doxycycline. The anti-anabolic action of the tetracyclines may cause an increase in blood urea. Studies to date indicate that this anti-anabolic effect does not occur with the use of Medomycin in patients with impaired renal function.

Photosensitivity: Photosensitivity manifested by an exaggerated sunburn reaction has been observed in some individuals taking tetracyclines, including doxycycline. Patients likely to be exposed to direct sunlight or ultraviolet light should be advised that this reaction can occur with tetracycline drugs and treatment should be discontinued at the first evidence of skin erythema.

Microbiological overgrowth: The use of antibiotics may occasionally result in overgrowth of non-susceptible organisms. Constant observation of the patient is essential. If a resistant organism appears, the antibiotic should be discontinued and appropriate therapy instituted.

Pseudomembranous colitis has been reported with nearly all antibacterial agents, including doxycycline, and has ranged in severity from mild to life-threatening. It is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of antibacterial agents.

Clostridium difficile associated diarrhoea (CDAD) has been reported with use of nearly all antibacterial agents, including doxycycline, and may range in severity from mild diarrhoea to fatal colitis. Treatment with antibacterial agents alters the normal flora of the colon leading to overgrowth of *C. difficile*.

C. difficile produces toxins A and B which contribute to the development of CDAD.

Hypertoxin producing strains of *C. difficile* cause increased morbidity and mortality, as these infections can be refractory to antimicrobial therapy and may require colectomy. CDAD must be considered in all patients who present with diarrhoea following antibiotic use. Careful medical history is necessary since CDAD has been reported to occur over two months after the administration of antibacterial agents.

Oesophagitis: Cases of oesophageal injuries (oesophagitis and ulceration), sometimes serious, have been reported.

Patients should be instructed to take doxycycline capsules with plenty of water (at least 100ml), remain upright and not take their treatment before going to bed (see section 4.2). Withdrawal of doxycycline and investigation of oesophageal disorder should be considered if symptoms such as dyspepsia or retrosternal pain occur. Caution is required in the treatment of patients with known oesophageal reflux disorders.

Bulging fontanelles: in infants and benign intracranial hypertension in juveniles and adults have been reported in individuals receiving full therapeutic dosages. These conditions disappeared rapidly when the drug was discontinued.

Venereal disease: When treating venereal disease, where co-existent syphilis is suspected, proper diagnostic procedures including dark-field examinations, should be utilised. In all such cases monthly serological tests should be made for at least four months.

Beta-haemolytic streptococci infections: Infections due to group A beta-haemolytic streptococci should be treated for at least 10 days.

Some patients with spirochete infections may experience a Jarisch-Herxheimer reaction shortly after doxycycline treatment is started. Patients should be reassured that his is a usually self-limiting consequence of antibiotic treatment of spirochete infections.

Medomycin contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

4.5. Interactions with other medicinal products and other forms of interaction

There have been reports of prolonged prothrombin time in patients taking warfarin and doxycycline. Because the tetracyclines have been shown to depress plasma prothrombin activity, patients who are on anticoagulant therapy may require downward adjustment of their anticoagulant dosage.

Since bacteriostatic drugs may interfere with the bactericidal action of penicillin, it is advisable to avoid giving Medomycin in conjunction with penicillin.

The absorption of doxycycline is impaired by concurrently administered antacids containing aluminium, calcium, magnesium or other drugs containing these cations; oral zinc, iron salts or bismuth preparations.

The serum half-life of doxycycline is shortened when patients are concurrently receiving alcohol, barbiturates, carbamazepine or phenytoin.

A few cases of pregnancy or breakthrough bleeding have been attributed to the concurrent use of tetracyclines with oral contraceptives.

The concurrent use of tetracyclines and methoxyflurane has been reported to result in fatal renal toxicity.

Laboratory test interactions

False elevations of urinary catecholamine levels may occur due to interference with the fluorescence test.

4.6. Fertility, pregnancy and lactation

<u>Pregnancy</u>

Doxycycline has not been studied in pregnant patients. It should not be used in pregnancy unless, in the judgement of the physician, it is essential for the welfare of the patient. (See section 4.3 about use during tooth development).

<u>Breast-feeding</u>

Tetracyclines are present in the milk of lactating women who are taking a drug of this kind and should therefore not be used in nursing mothers (See section 4.3 about use during tooth development).

4.7. Effects on ability to drive and use machines

The effect of doxycycline on the ability to drive or operate heavy machinery has not been studied. There is no evidence to suggest that doxycycline may affect these abilities.

4.8. Undesirable effects

The following adverse reactions have been observed in patients receiving tetracyclines, including doxycycline.

Adverse Reactions Table

System Organ Class	Very Common	Common	Uncommon	Rare	Frequency not known
Blood and lymphatic system disorders				Thrombocytopaenia Haemolytic anaemia Neutropenia Eosinophilia	
Immune system disorders		Anaphylactic Reaction (including Hypersensitivity, Henoch-Schonlein Purpura,		Drug Rash with Eosinophilia and Systemic Symptoms (DRESS)	Jarisch-Herxheimer reaction (see Section Warnings & Precautions)

System Organ Class	Very Common	Common	Uncommon	Rare	Frequency not known
		hypotension, Pericarditis, Angioedema, Exacerbation of systemic lupus erythematosus, Dyspnoea, Serum sickness, Peripheral oedema, Tachycardia and Urticaria)			
Endocrine disorders				Brown-black microscopic discoloration of thyroid glands	
Metabolism and nutrition disorders				Anorexia	
Nervous system disorders		Headache		Fontanelle bulging Benign intracranial hypertension	
Ear and labyrinth disorders				Tinnitus	
Vascular disorders				Flushing	
Gastrointestinal disorders		Nausea/vomiting	Dyspepsia (Heartburn/ gastritis)	Pseudomembranous colitis *C.difficile* diarrhoea Oesophageal ulcerations Oesophagitis Enterocolitis Inflammatory lesions (with monilial overgrowth) in the anogenital region Abdominal pain Diarrhoea Dysphagia Glossitis	
Hepatobiliary disorders				Hepatotoxicity Hepatitis Hepatic Function Abnormal	
Skin and subcutaneous tissue disorders	Photosensitivity Reaction	Rash including maculopapular and erythematous rashes		Toxic Epidermal Necrolysis Stevens-Johnson syndrome Erythema multiforme Exfoliative dermatitis Photoonycholysis Fixed eruption	
Musculoskeletal and connective tissue disorders				Arthralgia Myalgia	

System Organ Class	Very Common	Common	Uncommon	Rare	Frequency not known
Renal and urinary disorders				Blood Urea Increased	

4.9. Overdose

Management

Acute overdosage with antibiotics is rare. In the event of overdosage discontinue medication. Gastric lavage plus appropriate supportive treatment is indicated.

Dialysis does not alter serum half-life and thus would not be of benefit in treating cases of overdosage.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: Tetracyclines, ATC Code: J01 AA02.

Doxycycline is primarily bacteriostatic and is believed to exert its antimicrobial effect by the inhibition of protein synthesis. Doxycycline is active against a wide range of Gram-positive and Gram-negative bacteria and certain other micro-organisms. Doxycycline has a high degree of lipid solubility and a low affinity for calcium. It is highly stable in normal human serum. Medomycin will not degrade into an epianhydro form.

5.2. Pharmacokinetic properties

Doxycycline hydrochloride is readily absorbed from the gastro intestinal tract and 2 hours after a 200mg dose plasma concentrations of 2.6ug per ml falling to 1.45ug per ml at 24 hours have been reported. Following repeated doses of 100mg daily, concentrations of about 2g per ml are maintained. Similar values are reported after intravenous infusions.

Absorption is not significantly affected by the presence of food in the stomach or duodenum. Up to 90% of doxycycline in the circulation is reported to be bound to plasma proteins. Its biological half-life varies from about 15 hours after a single dose to 22 hour after repeated doses. Doxycycline is more lipid-soluble than tetracycline. It is widely distributed in body tissues and fluids but only low concentrations are achieved in cerebrospinal fluid. Doxycycline is slowly excreted mainly in the urine. It is stated not to accumulate in patients with renal impairment although excretion in the urine is reduced; doxycycline may be excreted directly into the gastro-intestinal tract in these patients. Nevertheless there have been reports of some accumulation in renal failure. There is also some inactivation in the liver but up to 25% of a dose has been excreted in the faeces in an active form.

5.3 Preclinical safety data

Not applicable.

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

Lactose monohydrate, starch maize pregelatinised, magnesium stearate. Capsule shells contain gelatin, titanium dioxide (E171), iron oxide Yellow (E172), quinoline yellow (E104), brilliant blue (E133) and erythrosine (E127).

6.2. Incompatibilities

None known.

6.3. Shelf Life

4 years.

6.4. Special precautions for storage

Store 30^{o}C in the original package, in order to protect form light and moisture.

6.5. Nature and contents of container

PVC-Al blisters in cartons of 100 and 1000 capsules. Not all pack sizes may be marketed.

6.6. Special precaution for disposal

No special requirements

7. MANUFACTURER

Medochemie (Far East) Ltd. (Oral Facility), No.40 Street 6 Vietnam Singapore Industrial Park II, Binh Duong Industry Service Urban Complex, Hoa Phu ward, Thu Dau Mot city, Vietnam.

8. PRODUCT REGISTRATION HOLDER

KOMEDIC SDN BHD, No. 4 Jalan PJS 11/14, Bandar Sunway, 46150 Petaling Jaya, Selangor, MALAYSIA

9. PRODUCT REGISTRATION NUMBER

MAL 19860568AZ

10. DATE OF FIRST AUTHORISATION/RENEWAL OF THE AUTHORISATION

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09/2025